

GEOMETRICAL ROTATIONS & STATE SUPERPOSITION ON THE BLOCH SPHERE

A visual beginner tutorial for single-qubit states, superposition, rotations, matrices, and measurement

Learning goal: Understand how a single qubit is represented, how superposition appears on the Bloch Sphere, how gates transform the state, and how measurement produces a classical result.

01 From Classical Bits to Qubits

A classical computer stores information using bits. A bit has one definite value at a time: 0 or 1. Quantum computers use qubits, whose state can contain contributions from both computational-basis states.

$$|0\rangle = [1, 0]^T \quad |1\rangle = [0, 1]^T$$

A general single-qubit state is written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Here α and β are generally complex probability amplitudes. They are not probabilities themselves.

02 From Amplitudes to Probabilities

$$P(0) = |\alpha|^2 \quad P(1) = |\beta|^2$$

A valid state must be normalized:

$$|\alpha|^2 + |\beta|^2 = 1$$

03 What Is Superposition?

If both amplitudes are non-zero, the qubit is in a superposition of $|0\rangle$ and $|1\rangle$. A standard example is:

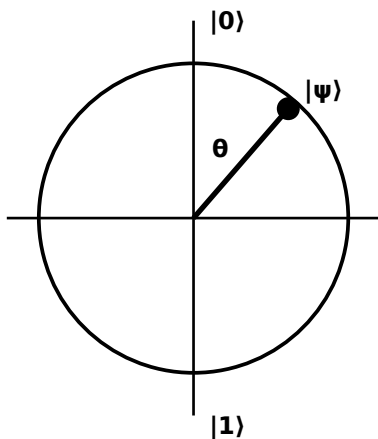
$$|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

For this state, the squared magnitudes of both amplitudes are $1/2$. Therefore, a computational-basis measurement gives 0 or 1 with equal probability.

Remember: Superposition describes the quantum state before measurement. A measurement produces one definite classical outcome.

04 THE BLOCH SPHERE

The Bloch Sphere is a geometric way to visualize the state of a single-qubit pure state. A normalized state can be represented by a point on the surface of a unit sphere.



BLOCH SPHERE

A point on the surface represents a pure qubit state.

θ = polar angle

ϕ = azimuthal angle

Gates move or rotate the point.

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi} \sin(\theta/2)|1\rangle$$

θ (theta) is the polar angle measured down from the north pole. ϕ (phi) is the azimuthal angle around the vertical axis.

05 Important Points

State	Location	Interpretation
$ 0\rangle$	North pole	Definite computational-basis 0
$ 1\rangle$	South pole	Definite computational-basis 1
$ +\rangle$	Equator	Equal-amplitude superposition
$ -\rangle$	Opposite equator point	Different relative phase

06 Why Geometry Helps

The Bloch Sphere connects algebra with visual intuition. When a gate changes a qubit state, its corresponding point on the sphere changes position. For many single-qubit gates, that change can be understood as a rotation around an axis.

Two views of the same state: Algebra $\rightarrow$ vectors and matrices Geometry $\rightarrow$ a point moving on the Bloch Sphere.

07 QUANTUM GATES AS TRANSFORMATIONS

A quantum gate transforms a qubit state. For a single qubit, the gate is represented by a 2×2 matrix.

$$|\psi'\rangle = U|\psi\rangle$$

The matrix U acts on the state vector $|\psi\rangle$ and produces the new state $|\psi'\rangle$.

08 Worked Example: Pauli-X

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Starting from $|0\rangle$:

$$X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

So X flips $|0\rangle$ to $|1\rangle$ and $|1\rangle$ to $|0\rangle$. On the Bloch Sphere, X corresponds to a π rotation about the x-axis.

09 Common Single-Qubit Gates

X	Y	Z	H
$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Bit flip π around x	π around y	Phase flip π around z	Creates/changes superposition

10 Phase Gates

The S and T gates change the relative phase of the $|1\rangle$ component. They are useful for understanding movement around the z-axis.

S	$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$	Adds a $\pi/2$ relative phase
T	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$	Adds a $\pi/4$ relative phase

Core connection: The matrix gives the exact algebraic transformation; the Bloch Sphere gives the geometric interpretation.

11 ARBITRARY ROTATIONS: Rx, Ry, Rz

Rotation gates describe rotations by an arbitrary angle. The angle controls how far the state rotates around the selected axis.

$$R_x(\theta) = \cos(\theta/2)I - i \sin(\theta/2)X$$

$$R_y(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$$

12 Worked Example: H Creates Superposition

Start with $|0\rangle$ and apply the Hadamard gate:

$$H = (1/\sqrt{2}) \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H|0\rangle = (1/\sqrt{2}) \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = (1/\sqrt{2}) \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Therefore:

$$H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2} = |+\rangle$$

The result is an equal superposition. Measuring in the computational basis gives 0 with probability 1/2 and 1 with probability 1/2.

13 From the Mathematics to Python

A raw NumPy simulator can represent a state as a vector and a gate as a matrix. The essential operation is matrix-vector multiplication.

```
import numpy as np
ket0 = np.array([1, 0], dtype=complex)
H = (1/np.sqrt(2))*np.array([[1,1],[1,-1]], dtype=complex)
state = np.dot(H, ket0)
print(state)
```

Expected output: approximately [0.7071, 0.7071]. The squared magnitudes are approximately 0.5 and 0.5.

Debugging checklist: Compare the program with the hand calculation. If they differ, check matrix entries, signs, complex dtype, normalization, dimensions, and multiplication order.

14 MEASUREMENT & STATE COLLAPSE

Before measurement, a qubit may be in a superposition. When measured in the computational basis, the result is a classical 0 or 1.

For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$:

$$P(0) = |\alpha|^2 \quad P(1) = |\beta|^2$$

If the outcome is 0, the post-measurement state is $|0\rangle$. If the outcome is 1, the post-measurement state is $|1\rangle$. This change to a definite outcome is commonly called state collapse.

15 Example: Measuring $|+\rangle$

$$|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$$

Outcome	Probability	Post-measurement state
0	$1/2 = 50\%$	$ 0\rangle$
1	$1/2 = 50\%$	$ 1\rangle$

16 The Complete Picture

Start $\rightarrow$ Transform $\rightarrow$ New state $\rightarrow$ Measure

$|0\rangle \rightarrow H \rightarrow |+\rangle \rightarrow \text{measurement} \rightarrow 0 \text{ or } 1$

17 Try It Yourself

- Write $|0\rangle$ and $|1\rangle$ as column vectors and verify normalization.
- Calculate $X|0\rangle$ and $X|1\rangle$ by hand.
- Apply H to $|0\rangle$ and calculate both measurement probabilities.
- Choose α and β satisfying $|\alpha|^2 + |\beta|^2 = 1$, then calculate $P(0)$ and $P(1)$.
- Implement one gate with NumPy and compare its output with your hand calculation.

18 Final Takeaways

Qubit	A normalized state vector with complex amplitudes.
Superposition	A state containing amplitudes for multiple basis states.
Bloch Sphere	A geometric visualization of a single-qubit pure state.
Gate	A transformation represented algebraically by a matrix.
Measurement	Produces a classical outcome according to the state's probabilities.

Final idea: Vectors and matrices tell us how a qubit changes; the Bloch Sphere helps us see that change.